Brownian clock mixtures: exact transforms, singular time evolution and the limits of one-time information

Project Stardust
A classical formalization companion

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Abstract

Two-sided Brownian motion evaluated at a signed Brownian clock and Brownian motion evaluated at its reflected clock have the same one-time Gaussian mixture laws. We reconstruct their exact characteristic functions, the boundary term in their fourth-order time equation, and the square-root cusp that prevents a finite initial time derivative at any nonzero Fourier frequency. The laws are genuine probabilities, but do not form a convolution semigroup. A separate covariance calculation proves the reflected-clock observation is not a martingale in its observed past. A hidden-clock local-time formula concerns the scenery average and does not transfer to the unaveraged observation. For arbitrary fixed even nonnegative compact smoothing profiles, an exact Gaussian Stein decomposition recovers the original curvature boundary correction in ordinary clock L^2, independently of the profile. Completed Fourier source means also retain the original hidden-clock Itô integral under explicit absolute and fourth-frequency summability. All proofs retain the standard Brownian variance convention and are matched to a frozen Lean library. The Brownian-time PDE mechanism is classical, with a direct primary comparison to Allouba and Zheng; no new PDE, fractional generator, full observed-filtration calculus, or priority claim is made.

1 The two actual models and what is being compared

Let W be a standard Brownian motion on $[0,\infty)$ and let B be an independent continuous two-sided Brownian scenery, normalized by

$$\mathbb{E}B(r) = 0, \qquad \mathbb{E}B(r)B(q) = \frac{|r| + |q| - |r-q|}{2}. \tag{1}$$

In particular $B(r)$ has variance $|r|$. On each half-line the scenery has the usual Brownian covariance, and the two halves have independent Gaussian finite-dimensional laws. Define

$$Z_t^- = B(W_t), \qquad Z_t^+ = B(|W_t|).$$

The superscripts label signed and reflected clocks; they are not positive and negative parts of a single random variable. Both processes are constructed on the product of the independent source spaces. Evaluation of continuous paths at a continuous random time gives continuous sample paths. Joint measurability follows from the jointly continuous evaluation map on continuous-path space and the measurability of the two source paths. Thus the expectations below refer to actual measurable compositions, not formal random times.

We prove results about their common one-time law μ_t, and state separately which assertions require the actual temporal coupling. Equality of one-time laws supplies neither equality of process laws nor equality of natural filtrations. The standard Brownian convention here is $\mathbb{E}W_t^2 = t$, with spatial generator $\frac{1}{2}\partial_x^2$. It differs from the variance-$2t$ convention used for the signed higher-heat kernels in another companion.

The exact source construction is in `SignedIBMRealization`. Its conditional Gaussian theorem retains the entire fixed clock path; the one-time comparison is in `IBMOneTimeMixture`. No change to either model or its verified hypotheses is made here.

2 The common Gaussian mixture

Write $\gamma_v = N(0, v)$, including $\gamma_0 = \delta_0$. The map $r \mapsto \gamma_{|r|}$ is a probability kernel. For example, for bounded continuous f, its integral is $\mathbb{E}f(\sqrt{|r|}G)$ with $G \sim N(0,1)$; dominated convergence gives continuity in r. The usual generating-class argument then gives measurability of the kernel on Borel sets. Define

$$\mu_t(A) = \int_{\mathbb{R}} \gamma_{|r|}(A)\, \gamma_t(dr). \tag{2}$$

This is a probability measure, since each conditional law has total mass one and the mixing law is a probability.

Theorem 2.1. *For every fixed $t \geq 0$, both Z_t^- and Z_t^+ have law μ_t. For every bounded Borel f,*

$$\mathbb{E}f(Z_t^-) = \mathbb{E}f(Z_t^+) = \int_{\mathbb{R}} \int_{\mathbb{R}} f(x)\, \gamma_{|r|}(dx)\, \gamma_t(dr). \tag{3}$$

The same formula holds for nonnegative Borel tests, with possibly infinite values, and for absolutely integrable tests.

Proof. Fix the clock source. At its value $W_t = r$, the signed observation is the scenery coordinate $B(r)$, whose law is $N(0, |r|)$ by (1). The reflected observation is $B(|r|)$, with the same variance and law. Independence makes the scenery source measure unchanged when the clock is fixed. Apply Tonelli first to indicators of Borel sets. The resulting integral is exactly (2), proving the two pushforward identities. Simple approximation proves the nonnegative test formula. Applying it to the absolute value, and then to positive and negative parts or real and imaginary parts, proves every integrable version. In particular bounded tests are integrable under these probability laws. □

There is a concrete limitation to this comparison. At fixed signed outer times $1, -1$, the scenery covariance is zero, whereas at the reflected times $|1|, |-1|$ it is one. More generally, conditional on a clock path w, the signed two-time covariance is

$$\frac{|w(s)| + |w(t)| - |w(t) - w(s)|}{2},$$

while the reflected covariance is $\min(|w(s)|, |w(t)|)$. The formulas follow directly from (1); the latter uses $a + b - |a - b| = 2\min(a, b)$ for nonnegative a, b. These are conditional scenery statements, not a claimed identification of the unconditional two-time laws from one-dimensional marginals.

Moments and a one-time scaling check

Let G_1, G_2 be independent standard normals on an auxiliary product space. The mixture construction gives the distributional identity

$$Z_t^{\pm} \overset{\text{law}}{=} t^{1/4}\sqrt{|G_2|}\, G_1.$$

Indeed $W_t \overset{\text{law}}{=} \sqrt{t}\, G_2$, and conditional scenery variance is its absolute value. This identity is about each fixed time. No simultaneous pathwise equality with this auxiliary pair is asserted. Gaussian absolute moments are finite of every fixed positive order, so Tonelli proves integrability before any moment factorization. Using $\mathbb{E}|G| = \sqrt{2/\pi}$, $\mathbb{E}G^2 = 1$, $\mathbb{E}G^4 = 3$, and $\mathbb{E}G^8 = 105$ gives

$$\mathbb{E}Z_t^{\pm} = 0, \qquad \mathbb{E}(Z_t^{\pm})^2 = \sqrt{\frac{2t}{\pi}}, \qquad \mathbb{E}(Z_t^{\pm})^4 = 3t, \qquad \mathbb{E}(Z_t^{\pm})^8 = 315t^2. \tag{4}$$

For example the eighth moment is $105\mathbb{E}W_t^4 = 315t^2$; all conditional integrands are nonnegative. The normal moment values follow by integration by parts against $e^{-x^2/2}$, with boundary terms zero; the absolute first moment follows by integrating $xe^{-x^2/2}$ on the positive half-line. For $t > 0$ the kurtosis is $3\pi/2 > 3$, so this nondegenerate one-time law is not Gaussian. The displayed global moment evaluations and this diagnostic are human consequences of the actual mixture; the formal actual quartic and eighth-integrability constructions are retained in the source map.

3 An exact transform and its boundary identity

Put $c = (2\pi)^{-1/2}$ and

$$L(b) = \mathbb{E}e^{-b|G|} = 2c\int_0^{\infty} e^{-r^2/2-br}\, dr, \qquad b \in \mathbb{R}.$$

The integral is finite for every fixed b: completing the square leaves a translated Gaussian tail. In the physical applications below $b \geq 0$, so $0 < L(b) \leq 1$ and $L(0) = 1$.

Proposition 3.1. *For angular Fourier frequency $\xi \in \mathbb{R}$,*

$$\widehat{\mu}_t(\xi) := \mathbb{E}e^{i\xi Z_t^{\pm}} = L\left(\frac{\xi^2\sqrt{t}}{2}\right). \tag{5}$$

For $b \geq 0$,

$$\mathbb{E}\{|G|e^{-b|G|}\} = 2c - bL(b), \qquad L'(b) = bL(b) - 2c. \tag{6}$$

The derivative at $b = 0$ is an ordinary two-sided parameter derivative.

Proof. Conditional on the clock, the characteristic function of the scenery normal is $e^{-\xi^2|W_t|/2}$. Its modulus is at most one, so Fubini is justified directly on the source probability product. Brownian scaling $|W_t| \overset{\text{law}}{=} \sqrt{t}|G|$ gives (5).

For the boundary identity set $h_b(r) = e^{-r^2/2-br}$ on $r \geq 0$. It has derivative $h_b' = -(r+b)h_b$, value $h_b(0) = 1$, and limit zero at infinity. Both h_b and rh_b are integrable for $b \geq 0$, by domination by a Gaussian and its first moment. Integration of the derivative on $[0, R]$ and then $R \to \infty$ yields

$$\int_0^{\infty} rh_b(r)\, dr = 1 - b\int_0^{\infty} h_b(r)\, dr.$$

Multiply by $2c$. This is the first formula in (6). Differentiating the probability integral for L gives the negative of this moment. On a neighborhood inside $b > 0$, the derivative integrands are dominated by $|G|$. Around zero, use instead $|G|e^{|G|}$ for $|b| \le 1$. This has finite Gaussian integral: its tail is a polynomial times $e^{-(|r|-1)^2/2}$ times a constant. These majorants justify differentiation, including at zero. □

Completing the square also gives the useful exact expression

$$L(b) = 2ce^{b^2/2} \int_b^\infty e^{-v^2/2}\, dv.$$

This identity is a human re-expression of the defining integral; it is not a separate mapped special-function implementation. For $b > 0$, the integral representation proves the elementary upper bound $L(b) \le 2c/b$ by dropping $e^{-r^2/2}$ and integrating e^{-br}. The bound $L(b) \le 1$ remains preferable near zero.

4 The singular fourth-order time equation

For $a \ge 0$ define $H_a(t) = L(a\sqrt{t})$ for $t \ge 0$. This is continuous at zero and is positive and at most one. For $t > 0$, the chain rule and (6) give

$$H_a'(t) = \frac{a^2}{2} H_a(t) - \frac{ac}{\sqrt{t}}. \tag{7}$$

At $a = 0$ this is the constant identity. At $a > 0$ the formula follows from $\partial_t \sqrt{t} = 1/(2\sqrt{t})$ and $aL'(a\sqrt{t}) = a^2\sqrt{t}L(a\sqrt{t}) - 2ac$.

Theorem 4.1. *At every $t > 0$ and every real frequency,*

$$\partial_t \widehat{\mu}_t(\xi) = \frac{\xi^4}{8} \widehat{\mu}_t(\xi) - \frac{\xi^2}{2\sqrt{2\pi t}}. \tag{8}$$

Equivalently, for $T \ge 0$,

$$\widehat{\mu}_T(\xi) = 1 + \frac{\xi^4}{8} \int_0^T \widehat{\mu}_s(\xi)\, ds - \xi^2 c\sqrt{T}. \tag{9}$$

Proof. Substitute $a = \xi^2/2$ into (7) and use (5). To integrate to zero, first apply the ordinary fundamental theorem on $[\varepsilon, T]$. The bounded characteristic function is integrable in time and $s^{-1/2}$ is integrable at zero. Letting $\varepsilon \downarrow 0$ and using $\widehat{\mu}_0(\xi) = 1$ proves (9), since $\int_0^T s^{-1/2} ds = 2\sqrt{T}$. The $T = 0$ identity is immediate. □

The sign of the fourth-order term is positive in Fourier space. A stable biharmonic evolution instead has multiplier $e^{-t\xi^4}$. Here the negative source in (8), depending on the initial law, is indispensable. Omitting it does not preserve the actual probability characteristic function. In fact the derivative is strictly negative for $a > 0$:

$$H_a'(t) = -\frac{a}{2\sqrt{t}} \mathbb{E}\{|G| e^{-a\sqrt{t}|G|}\} < 0.$$

Strictness follows because $|G| > 0$ almost surely and the integrand is integrable and positive there. Thus the positive fourth-order term and the negative boundary term compensate in an exact probability identity; the fourth-order term by itself is not its Markov generator.

A concrete spatial formulation with all interchanges checked

Let $\phi \in L^1([-R,R];\mathbb{C})$ for finite $R \geq 0$, extended by zero, and set

$$f(x) = \int_{-R}^{R} \phi(\xi) e^{i\xi x}\, d\xi, \qquad u(t,x) = \mathbb{E} f(x + Z_t^{\pm}).$$

Then f is bounded and smooth, with every spatial derivative obtained by inserting $(i\xi)^k$ under the integral. In particular $\|f^{(k)}\|_\infty \leq R^k \|\phi\|_1$ for $k > 0$. Fubini uses the integrable majorant $|\phi(\xi)|$ and gives

$$u(t,x) = \int_{-R}^{R} \phi(\xi) e^{i\xi x} \widehat{\mu}_t(\xi)\, d\xi.$$

On a compact positive time interval $t \geq \delta > 0$, the absolute time derivative of the multiplier is at most $R^4/8 + R^2 c/(2\sqrt{\delta})$ by (8). This constant times $|\phi|$ dominates time difference quotients locally, and $R^4|\phi|$ dominates the fourth spatial derivative. Consequently dominated differentiation proves

$$\partial_t u(t,x) = \tfrac{1}{8}\partial_x^4 u(t,x) + \frac{f''(x)}{\sqrt{8\pi t}}, \qquad u(0,x) = f(x). \tag{10}$$

Indeed $(i\xi)^4 = \xi^4$ and $(i\xi)^2 = -\xi^2$, giving both the sign and the constant. Initial convergence is uniform in x: dominate $|\phi(\xi)|\,|\widehat{\mu}_t(\xi) - 1|$ by $2|\phi|$. This compact-frequency spatial statement is an explicit human Fubini consequence of the formal characteristic equation. It is not an asserted general generator-domain theorem or a pathwise Itô formula. Its PDE mechanism is the classical Brownian-time equation of Allouba–Zheng [1], with the same Brownian normalization.

5 The square-root cusp and failure of convolution composition

The parameter derivative at zero yields

$$\lim_{t \downarrow 0} \frac{H_a(t) - 1}{\sqrt{t}} = -2ca. \tag{11}$$

This follows by substituting $b = a\sqrt{t}$ into the defining difference quotient of $L'(0)$; $a = 0$ is constant. For $a > 0$, the limit is strictly negative, so H_a cannot have a finite right time derivative at zero. If it had derivative d, then $(H_a(t) - 1)/\sqrt{t} = ((H_a(t) - 1)/t)\sqrt{t} \to 0$, contradicting (11). The formal source also proves this by composing a hypothetical time derivative with $t = s^2$: the chain rule gives derivative zero, whereas $L(as)$ has derivative $-2ca$. Thus the cusp conclusion does not rest on a formal interchange of a singular derivative at the origin.

Proposition 5.1. *The probability family μ_t is not a convolution semigroup. In particular it cannot be the marginal family at zero of a process with stationary independent increments.*

Proof. Suppose $\mu_{s+t} = \mu_s * \mu_t$ for all nonnegative s, t. Convolution multiplies characteristic functions, so at $\xi = 1$ and $t = 1$ we would have

$$H_{1/2}(s+1) = H_{1/2}(s) H_{1/2}(1), \qquad s \geq 0.$$

The last factor is strictly positive. Therefore $H_{1/2}(s) = H_{1/2}(s+1)/H_{1/2}(1)$ would have a finite right derivative at zero, since the numerator is differentiable at positive time one. This contradicts the cusp. For stationary independent increments starting at zero, the law at $s+t$ is the convolution of the law at s with that of the independent length-t increment, giving precisely the excluded law. □

There is a second explicit check using (4). The centered law $\mu_1 * \mu_1$ has second moment $2\sqrt{2/\pi}$, whereas μ_2 has second moment $2/\sqrt{\pi}$. Both moments are finite, and independent centered summands add their variances, so the measures differ. This particular witness is a human deduction from the actual mixture moments; the formal convolution obstruction uses the preceding transform argument. Neither argument excludes a general nontranslation Markov model merely from its one-time marginals. Convolution composition and the full Markov property have different hypotheses.

6 A temporal obstruction requiring the actual reflected model

We now use $Z_t = Z_t^+$ on its original source product, rather than only the family μ_t. All pairings are integrable by (4) and Cauchy–Schwarz. Conditional on the clock, (1) at nonnegative times gives

$$\mathbb{E}_B\{Z_s(Z_t - Z_s)\} = \min(|W_s|, |W_t|) - |W_s| = -(|W_s| - |W_t|)_+.$$

Fubini is justified by the original joint L^2 bounds. Hence

$$\mathbb{E}\{Z_s(Z_t - Z_s)\} = -\mathbb{E}(|W_s| - |W_t|)_+. \tag{12}$$

The nonnegative right-hand integrand is bounded by $|W_s|$, so is integrable. At $s = 1, t = 2$ its expectation is strictly positive. Indeed the event

$$1 < W_1 < 2, \qquad -1 < W_2 - W_1 < -\tfrac{1}{2}$$

has positive probability: its two coordinates are independent nondegenerate standard normals, each interval has positive Gaussian mass, and the event probability is their positive product. On it $0 < W_2 < W_1$, so $(|W_1| - |W_2|)_+ > 0$. Thus (12) is strictly negative at those times.

If Z were a square-integrable martingale for its raw observed past, then $\mathbb{E}(Z_2 - Z_1 \mid \sigma(Z_r : r \le 1)) = 0$. Pair with the past-measurable $Z_1 \in L^2$, or first with bounded truncations and pass by Cauchy–Schwarz, to get zero in (12), a contradiction. More generally, every closed past subspace containing Z_1 fails to project Z_2 onto Z_1: the orthogonal projection identity paired with Z_1 would give that same zero. This is the exact actual temporal obstruction; it does not claim that the process has no other observed martingales, or that a Markov property is disproved by the martingale test alone.

7 A source mean has a hidden-clock local-time formula

The conditional scenery average of a Fourier cosine is

$$M_t(\xi) = \mathbb{E}_B \cos(\xi B(|W_t|)) = e^{-a|W_t|}, \qquad a = \xi^2/2.$$

This is an actual scalar function of the Brownian clock. Its ordinary Brownian filtration and the observed IBM filtration are not identified by the one-time mixture theorem.

Use the symmetric local time convention fixed by Tanaka's formula

$$|W_t| = \int_0^t \operatorname{sgn}(W_s)\, dW_s + L_t^0, \qquad \operatorname{sgn}(0) = 0.$$

The original constructed Brownian mass path is this L^0; its continuous Stieltjes measure is supported where $W_s = 0$. Tanaka and that supported-mass construction are established in the sticky-clock companion [2]; here we use their stated classical Brownian identity.

For $a \geq 0$ a genuine global C^2 extension of $x \mapsto e^{-ax}$ from the nonnegative half-line is

$$F_a(x) = \begin{cases} e^{ax} - 2ax, & x < 0, \\ e^{-ax}, & x \geq 0. \end{cases}$$

Its one-sided values and first derivatives agree at zero, with $F_a(0) = 1$ and $F_a'(0) = -a$. Its second derivative is $a^2 e^{-a|x|}$ on both sides and continuously at zero. The derivative is bounded by $2a$ and the second derivative by a^2. Thus the ordinary C^2 semimartingale Itô theorem applies to $F_a(|W|)$ on each finite horizon. The quadratic variation of $\int \operatorname{sgn}(W_s) dW_s$ is t: the zero set has Lebesgue measure zero almost surely, by $\mathbb{E} \int_0^T 1_{\{W_s=0\}} ds = \int_0^T \mathbb{P}(W_s = 0) ds = 0$. The finite variation local time contributes no quadratic variation. The supported Stieltjes integral uses $F_a'(|W_s|) = -a$ when $dL_s^0 \neq 0$. Itô therefore gives, at every fixed T almost surely,

$$M_T(\xi) - 1 = -a \int_0^T \operatorname{sgn}(W_s) e^{-a|W_s|} \, dW_s - a L_T^0 + \frac{a^2}{2} \int_0^T M_s(\xi) \, ds. \tag{13}$$

The stochastic integrand is bounded by a and predictable in the clock filtration, so its integral is square integrable with mean zero. The time integral is bounded by T. Tanaka implies $L_T^0 = |W_T| - \int_0^T \operatorname{sgn}(W_s) dW_s \in L^2$, and $\mathbb{E} L_T^0 = \mathbb{E}|W_T| = 2c\sqrt{T}$. Taking expectations in (13) is consequently justified and gives exactly (9).

The boundary term of the one-time PDE has a concrete hidden-clock origin. This does not make (13) an Itô formula for the unaveraged $B(|W_t|)$ in its observed filtration. Its left side has already integrated out the scenery, and its stochastic integral is with respect to the original clock Brownian motion. The source theorem is almost sure for each fixed horizon; no simultaneous all-horizons version is claimed here.

8 A profile-independent boundary correction in ordinary clock L^2

The boundary in the preceding formula can also be recovered from the literal curvature of spatially smoothed scenery. We identify the residual in the original smoothing and prove convergence in ordinary clock L^2. Averaging over scenery remains essential throughout this section.

Fix a real Schwartz function g, a shift $a \in \mathbb{R}$, and an even, nonnegative C^2 function ρ of compact support with $\int_{\mathbb{R}} \rho = 1$. Choose $R > 0$ with support in $[-R, R]$. For every $\varepsilon > 0$ set

$$\rho_\varepsilon(u) = \varepsilon^{-1} \rho(u/\varepsilon), \qquad S_\varepsilon(r) = \int_{\mathbb{R}} \rho_\varepsilon(u) B(r-u) \, du, \qquad v_\varepsilon(r) = \mathbb{E}_B S_\varepsilon(r)^2.$$

The integral defining S_ε is a Bochner integral in the original scenery L^2. Compact support and continuity of $r \mapsto B(r)$ in that Hilbert norm prove its integrability. The result is a centered Gaussian variable: Riemann sums are Gaussian and converge in L^2, hence their characteristic functions converge to that of the centered Gaussian with the limiting variance. Derivatives in r are obtained by moving them to the compact C^2 kernel. Write S_ε' and S_ε'' for these genuine Gaussian L^2 derivatives. At each fixed r, their joint Gaussian moments of every finite order exist.

The variance curvature is the original kernel

The covariance (1) and two applications of Fubini give

$$v_\varepsilon(r) = \int \rho_\varepsilon(u) |r-u| \, du - \tfrac{1}{2} \iint \rho_\varepsilon(u) \rho_\varepsilon(w) |u-w| \, du \, dw. \tag{14}$$

All variables u, w range over compact supports; the covariance is bounded there for a fixed r, so the Fubini steps are absolute. The second term is independent of r. Splitting the first integral at $u = r$ and applying the fundamental theorem of calculus gives

$$v_\varepsilon'(r) = 2\int_{-\infty}^{r} \rho_\varepsilon(u)\,du - 1, \qquad v_\varepsilon''(r) = 2\rho_\varepsilon(r). \tag{15}$$

Thus $|v_\varepsilon'| \le 1$, v_ε' equals the sign of r when $|r| > \varepsilon R$, and v_ε is even. The curvature is twice the original profile, rather than its square or self-convolution.

Jensen's inequality in the scenery Hilbert space gives the uniform original-coupling estimate

$$\mathbb{E}_B|S_\varepsilon(r) - B(r)|^2 \le \int \rho_\varepsilon(u)\mathbb{E}_B|B(r-u) - B(r)|^2 du = \int \rho_\varepsilon(u)|u|du \le \varepsilon R. \tag{16}$$

In particular $v_\varepsilon(0) \le \varepsilon R$. The slope bound yields $v_\varepsilon(r) \le 2\varepsilon R$ when $|r| \le \varepsilon R$. Also (14) gives $|v_\varepsilon(r) - |r|| \le 2\varepsilon R$ for every r: the first-term error is at most εR and the absolute second term at most εR.

`IBMCompactVarianceGeometry.variance_second`.

`IBMCompactVarianceGeometry.slope_bound`.

`IBMDilatedMollifier.scenery_error`.

The literal Gaussian correction and its exact decomposition

Define the scalar source-curvature residual

$$C_\varepsilon(r) = \mathbb{E}_B\left[g''(a + S_\varepsilon(r))(S_\varepsilon'(r))^2 + g'(a + S_\varepsilon(r))S_\varepsilon''(r)\right] - \tfrac{1}{4}\mathbb{E}_B g^{(4)}(a + S_\varepsilon(r)). \tag{17}$$

These are actual integrals against the original scenery law. Bounded Schwartz derivatives and finite Gaussian second moments give absolute integrability of both terms inside the expectation. No conditional projection of a singular limiting field defines them.

For any centered jointly Gaussian triple (X, Y, V) and a smooth function with the bounded derivatives needed below, Gaussian integration by parts gives

$$\begin{aligned}\mathbb{E}[Y^2 h(X)] &= (\mathbb{E}Y^2)\mathbb{E}h(X) + (\mathbb{E}XY)^2\mathbb{E}h''(X),\\ \mathbb{E}[Vh(X)] &= (\mathbb{E}XV)\mathbb{E}h'(X).\end{aligned}$$

For completeness, realize the triple as linear forms in a finite standard Gaussian vector ξ. The elementary identity $\mathbb{E}[\xi_j F(\xi)] = \mathbb{E}[\partial_j F(\xi)]$ gives the second formula once. Apply it first to $Yh(X)$ and then to $h'(X)$ for the first formula. Gaussian moments dominate every term. The argument uses no inverse covariance matrix and includes degenerate triples.

Apply these identities with $X = S_\varepsilon$, $Y = S_\varepsilon'$ and $V = S_\varepsilon''$. Hilbert-space differentiation gives

$$v_\varepsilon' = 2\mathbb{E}_B(S_\varepsilon S_\varepsilon'), \qquad v_\varepsilon'' = 2\mathbb{E}_B\left[(S_\varepsilon')^2 + S_\varepsilon S_\varepsilon''\right].$$

Using (15), the literal residual is exactly

$$C_\varepsilon(r) = \rho_\varepsilon(r)A_{2,\varepsilon}(r) + \frac{(v_\varepsilon'(r))^2 - 1}{4}A_{4,\varepsilon}(r), \qquad A_{j,\varepsilon}(r) = \mathbb{E}_B g^{(j)}(a + S_\varepsilon(r)). \tag{18}$$

The constants $1/4$ and 2 come from the displayed actual covariance derivatives. They are not adjusted to fit a desired limit.

`IBMCorrectionStein.correction_identity`.

`IBMCompactOriginalBoundary.originalResidual_literal`.

`IBMCompactOriginalBoundary.originalResidual_identity`.

Constructing the occupation mass and its ordinary L^2 limit

Let L_T^0 denote the symmetric Tanaka local time of the original clock, with the normalization already used above,

$$L_T^0 = |W_T| - \int_0^T \operatorname{sgn}(W_s)\, dW_s.$$

Evenness of ρ makes the literal reflected occupation mass

$$M_\varepsilon(T) = \int_0^T \rho_\varepsilon(|W_s|)ds = \int_0^T \rho_\varepsilon(W_s)ds.$$

Ordinary C^2 Itô calculus applied to $v_\varepsilon(W)$ and (15) gives, for each fixed T,

$$M_\varepsilon(T) = v_\varepsilon(W_T) - v_\varepsilon(0) - \int_0^T v_\varepsilon'(W_s)\, dW_s. \tag{19}$$

The derivatives are bounded at each fixed width, and $|v_\varepsilon(W_T) - v_\varepsilon(0)| \le |W_T|$. Thus every term is in clock L^2, and the triangle inequality and Itô isometry give the useful human bound

$$\|M_\varepsilon(T)\|_2 \le 2\sqrt{T}. \tag{20}$$

The endpoint difference in (19) tends to $|W_T|$ in L^2, by the uniform variance estimate above. For every $s > 0$, $\mathbb{P}(W_s = 0) = 0$ by its Gaussian density; Tonelli therefore gives zero $ds\, d\mathbb{P}$ measure to $W_s = 0$ on $[0, T]$. At every nonzero value, v_ε' tends to the sign, and both are bounded by one. Dominated convergence in $L^2(ds\, d\mathbb{P})$ and Itô isometry prove

$$M_\varepsilon(T) \longrightarrow L_T^0 \quad \text{in clock } L^2 \quad (\varepsilon \downarrow 0). \tag{21}$$

This constructs the needed mass limit without assuming a spatial local-time density or a pathwise occupation formula. It holds through all positive widths, not only a selected sequence. One may verify this last assertion along every positive sequence converging to zero, which characterizes this limit in a metric space.

`IBMCompactClockMass.clockMass_identity`.

`IBMCompactClockMass.clockMass_reflected_ae`.

`IBMDilatedClockMassLimit.reflected_pathIntegral_continuum_tendsto`.

Theorem 8.1 (Original boundary correction)**.** *For every fixed $T \ge 0$ and every fixed profile satisfying the stated hypotheses, the literal original-law random functional*

$$H_\varepsilon(T) = \int_0^T C_\varepsilon(|W_s|)\, ds$$

belongs to clock L^2 and satisfies

$$H_\varepsilon(T) \longrightarrow g''(a)L_T^0 \quad \textit{in ordinary clock } L^2 \quad (\varepsilon \downarrow 0). \tag{22}$$

The limit is independent of the fixed admissible profile.

Proof. Put $K = \|g'''\|_\infty$ and $N = \|g^{(4)}\|_\infty$. On the support of $\rho_\varepsilon(r)$, the variance bound and Cauchy–Schwarz give

$$|A_{2,\varepsilon}(r) - g''(a)| \le K\mathbb{E}_B|S_\varepsilon(r)| \le K\sqrt{2\varepsilon R}.$$

Since $\rho_\varepsilon \geq 0$, the first term of (18), integrated along the reflected clock, differs from $g''(a)M_\varepsilon(T)$ by at most $K\sqrt{2\varepsilon R}\, M_\varepsilon(T)$ pointwise. Its L^2 error is at most $K\sqrt{2\varepsilon R}\, \|M_\varepsilon(T)\|_2$, which tends to zero by (20). Each such weighted integral is in L^2 by this estimate and the mass identity.

The second residual term is bounded in absolute value by $N/4$. It tends to zero for every nonzero r; in fact it vanishes when $|r| > \varepsilon R$. It is a continuous function of r, as follows from continuity of the Gaussian L^2 scenery curve and the Lipschitz property of $g^{(4)}$. Composition with the continuous clock is jointly measurable. The zero-occupation fact above and dominated convergence on the finite product measure $ds\, d\mathbb{P}$ show its time integral tends to zero in clock L^2: Cauchy–Schwarz bounds the squared integral by T times the integral of its square. It is also in L^2 at each width, since its absolute time integral is at most $NT/4$. Combine the two terms and (21) to obtain (22). The case $T = 0$ is immediate. □

`IBMCompactSourceAverage.sourceAverage_layer_error`.
`IBMCompactWeightedBoundary.weightedClass_continuum_tendsto`.
`IBMCompactBoundaryRemainder.integralClass_continuum_tendsto`.
`IBMCompactOriginalBoundary.originalPathIntegral_continuum_tendsto`.

The original residual, the whole positive-width limit and its ordinary clock L^2 topology are exactly the mapped theorem's scope. The explicit uniform estimate (20) is a human consequence of the displayed actual Itô identity and slope bound. The profile is fixed while its width shrinks: no uniformity over varying profiles with uncontrolled support radii is asserted. Evenness is used to replace W by $|W|$ in the occupation mass; positivity supplies the weighted error and slope bounds.

There is no extra factor $1/2$ in (21): the reflected integral equals the full even-kernel integral. If the curvature appears with the usual Itô prefactor $1/2$, the residual contribution is consequently $g''(a)L_T^0/2$. The subtraction in (17) concerns a *scenery expectation*. This theorem does not prove a joint scenery/clock L^2 limit of the unaveraged curvature, convergence of an intrinsic predictable IBM integrand, or a physical-time Itô formula for $g(a+Z_T^+)$. It recovers a precise original boundary correction with all of those other obligations kept distinct.

9 Completed Fourier means and the original integral

The single cosine formula in Section 7 extends to an infinite series without introducing a different stochastic integral. Let $b_k \in \mathbb{R}$ and $c_k \geq 0$, for $k \geq 0$, satisfy

$$B_0 = \sum_k |b_k| < \infty, \qquad B_2 = \sum_k |b_k| c_k^2 < \infty. \tag{23}$$

Then $B_1 = \sum_k |b_k| c_k < \infty$, since $c \leq 1 + c^2$. Write

$$m(r) = \sum_k b_k e^{-c_k r} \quad (r \geq 0), \qquad A = \sum_k b_k c_k, \qquad M_t = m(|W_t|).$$

The series defining m and its first two right derivatives converge uniformly on the half-line. In particular M is an actual bounded continuous clock process. No positivity of the coefficients is needed.

Theorem 9.1. *Under* (23)*, for each fixed $T \geq 0$, almost surely,*

$$M_T - \sum_k b_k = \int_0^T U_s\, dW_s - AL_T^0 + \frac{1}{2}\int_0^T \sum_k b_k c_k^2 e^{-c_k|W_s|}\, ds, \tag{24}$$

$$U_s = -\operatorname{sgn}(W_s) \sum_k b_k c_k e^{-c_k|W_s|}.$$

Here U is predictable in the Brownian clock filtration and belongs to $L^2(ds\,d\mathbb{P})$ on $[0,T]$. Its original finite sums

$$U_s^{(N)} = -\operatorname{sgn}(W_s)\sum_{k<N} b_k c_k e^{-c_k|W_s|}$$

converge to U in that space, and their ordinary Brownian Itô integrals converge to the displayed integral in terminal L^2.

Proof. Use the global extensions F_c of Section 7: $F_c(x) = e^{-cx}$ for $x \geq 0$ and $F_c(x) = e^{cx} - 2cx$ for $x < 0$. They are C^2 with $F_c(0) = 1$, $F_c'(0) = -c$,

$$|F_c'(x)| \leq 2c, \qquad F_c''(x) = c^2 e^{-c|x|}, \qquad |F_c''(x)| \leq c^2.$$

At the origin the two first derivatives both equal $-c$ and the two second derivatives both equal c^2, so the gluing is genuinely C^2. On $[-R,R]$, $|F_c(x)| \leq 1 + 2cR$. Thus $F = \sum_k b_k F_{c_k}$ converges locally uniformly, its first derivative series converges uniformly on all of $\mathbb{R}$ by $2B_1$, and its second derivative series converges uniformly by B_2. The fundamental theorem of calculus on each compact interval, applied successively to finite sums and their uniformly converging derivatives, proves $F \in C^2(\mathbb{R})$ with these derivatives. Consequently $F(r) = m(r)$ for $r \geq 0$, $F(0) = \sum b_k$, $F'(0) = -A$, and $F''(r) = \sum b_k c_k^2 e^{-c_k r}$ there.

`ReflectedDampingSeries.value_C2`.

`ReflectedDampingSeries.first_uniform`.

`ReflectedDampingSeries.second_uniform`.

For completeness, the reflected Itô step uses $G(x) = F(|x|) - F'(0)|x|$. Its two first derivatives at zero are zero, and its two second derivatives equal $F''(0)$; thus G is C^2, with $G'(x) = \operatorname{sgn}(x)(F'(|x|) - F'(0))$ and $G''(x) = F''(|x|)$. These derivatives are bounded. Apply ordinary Brownian C^2 Itô to $G(W)$, and add $F'(0)$ times Tanaka's formula. The martingale fields combine to $\operatorname{sgn}(W_s)F'(|W_s|) = U_s$ and the local-time term is $F'(0)L_T^0 = -AL_T^0$. The drift is bounded by $B_2/2$, and $|U_s| \leq B_1$. This proves (24) with all terms integrable. The value at $W_s = 0$ is immaterial to the time–probability class, because that set has zero product measure. Predictability also follows directly: the clock is a continuous adapted process, and composition with a Borel function preserves predictable measurability.

`GaussianReflectedC2Ito.reflected_ito_pathwise`.

`ReflectedDampingSeriesIto.actual_ito`.

`ReflectedDampingSeriesIto.running_actual`.

Finally the pointwise tail bound gives the explicit estimates

$$\|U - U^{(N)}\|_{L^2(ds\,d\mathbb{P})} \leq \sqrt{T}\sum_{k\geq N} |b_k| c_k, \qquad \left\| \int_0^T (U_s - U_s^{(N)})\, dW_s \right\|_2 = \|U - U^{(N)}\|_{L^2(ds\,d\mathbb{P})}. \tag{25}$$

The second equality is the isometry of the original Brownian integral. Hence the native finite sums identify exactly that integral, with no separately defined limiting integral.

`ReflectedDampingNativeIntegral.partial_tendsto`.

`ReflectedDampingNativeIntegral.integral_tendsto`. □

The explicit tail estimate (25) is a human consequence of the mapped bounds and isometry; the convergence and identity of the completed predictable field are named formal results. The theorem asserts equality almost surely at each fixed horizon. It does not add a simultaneous all-horizon version theorem.

An actual IBM source mean is included as follows. Let

$$f(x) = \sum_k w_k \cos(\beta_k + a_k x), \qquad \sum_k |w_k| < \infty, \qquad \sum_k |w_k| a_k^4 < \infty.$$

The series is uniformly convergent, so f is bounded and continuous. For every deterministic clock value r, Gaussian integration and the summable bound $|w_k|$ give

$$\mathbb{E}_B f(B(|r|)) = \sum_k w_k \cos\beta_k \, e^{-a_k^2|r|/2}.$$

The sine terms vanish by the centered Gaussian symmetry. Uniform convergence and continuity give the same actual source mean at $r = W_t$; equivalently it is the conditional expectation over the independent scenery source with the clock fixed. Take $b_k = w_k \cos\beta_k$ and $c_k = a_k^2/2$. Then $\sum |b_k| \le \sum |w_k|$ and $\sum |b_k| c_k^2 \le \frac{1}{4} \sum |w_k| a_k^4$, so Theorem 9.1 applies. In particular the local-time coefficient is $-\frac{1}{2} \sum_k w_k \cos\beta_k \, a_k^2$, with the same normalization as the single-frequency case.
`IBMFourierSourceMeanIto.mean_actual`.
`IBMFourierSourceMeanIto.actual_ito`.

This is a classical reflected Itô argument with an explicit completion and an exact formal correspondence. Its summability conditions are sufficient, not claimed necessary or optimal. Neither M_t nor its integrand is asserted measurable in the observed IBM past. Convergence in the original clock predictable space does not establish a representation theorem, stochastic integration theory or Itô formula for the unaveraged $f(Z_t^+)$.

10 Primary comparison and exact formal scope

Allouba and Zheng [1], Theorem 0.1, equations (0.1)–(0.2), already give the Brownian-time initial-data source and fourth-order PDE. Their Section 1, first part of the proof, obtains the source by integrating the reflected clock density by parts twice. We read that theorem statement and the complete Brownian-time part of its proof on printed pages 2 and 4–5 of the primary preprint. We do not import the excursion-limit appendix, exit-time theorems or their separate half-derivative conditional generator. In particular the phrase "half derivative" there is not a license to assert a Caputo equation or an observed-past representation theorem here.

The contributions of this companion are a readable reconstruction and an exact correspondence to verified source statements. The Gaussian mixture, Brownian-time PDE, normal moment tests, cusp and convolution test are classical mechanisms; no priority claim follows from their formalization or from grouping them in one paper. The compact-frequency PDE, integrated endpoint formula, explicit moment evaluations and particular second-moment convolution witness are labelled human consequences with proofs, not newly named Lean declarations. The primary literature already supplies a broader PDE setting than the compact-frequency illustration.

The frozen baseline is

`bab688bb6140cd6316c80680095a914ea0d346db`.

Principal source families, read in full, are `IBMOneTimeMixture` and `IBMOneTimeFourier`.
The boundary and time calculations use `IBMGaussianLaplaceBoundary`.
Their derivative chain is `IBMClockTransformDerivative` and `IBMClockTimeEvolution`.
The exact final characteristic equation is in `IBMFourierTimeEquation`.
The initial and convolution obstructions are in `IBMInitialCusp` and `IBMConvolutionObstruction`.
The actual backtracking test is in `IBMClockBacktracking`.

The past-projection consequence is in `IBMTemporalCovarianceObstruction`.
The source-mean and damped reflection identities are in `IBMOneTimeSourceMeanIto`.
The original C^2 extension is in `GaussianDampedReflectedIto`.

The accompanying map gives exact declaration headers, source line links and byte hashes; the baseline receipt checks the existing full build and transitive axiom audit. The editorial work changes no mathematical source and runs no redundant Lean build. Nothing here completes intrinsic observed-filtration IBM calculus, constructs a Markov semigroup from μ_t, establishes a full physical-time generator domain, or upgrades a one-time comparison to an equality of process laws. Those distinctions are mathematical hypotheses, not matters of notation.

References

[1] H. Allouba and W. Zheng, Brownian-time processes: the PDE connection and the half-derivative generator, *Ann. Probab.* **29** (2001), 1780–1795. Primary preprint arXiv:1005.3801v1, deposited 2010; manuscript dated January 2001. `https://arxiv.org/abs/1005.3801`.

[2] Project Stardust, *Brownian sticky clocks: construction, holding time and causal recovery in the observed filtration*, reviewed editorial companion, `manuscripts/sticky-clock-recovery/`. The Brownian local-time and supported-mass construction is the prerequisite used in Section 7.